



S I M A T S
ENGINEERING

DSA0405 – Fundamentals of Data Science

CO3-AT1-Case Study Solutions

(Case Study 6 – Case Study 10)

Name: Selvamagal M

Register Number: 192472013

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1. Introduction

Statistics is one of the basic tools of Data Science. Before building any model or making any decision from data, it is important to first understand simple things about the data, like where it is centred and how spread out it is. This assignment has five case studies based on descriptive statistics.

The case studies cover the z-score (Case Study 6), mean/median/quartiles/IQR and outlier detection (Case Study 7 and Case Study 9), and variance/standard deviation (Case Study 8 and Case Study 10). Case Study 9 also looks at skewness. Each case study is solved step by step below, with the formula, the calculation, and a short explanation of what the result means.

2. Case Study 6 – Detecting an Unusual Customer

Problem Statement

An online shopping website has an average customer spend of ₹2,000 with a standard deviation of ₹400. One customer spent ₹3,200. We need to find how unusual this spending is using the z-score.

Objective: To calculate the z-score for the given customer and use it to check whether the spending amount is unusually high, and to understand what a negative z-score would mean in this same situation.

Given Data

Parameter	Value
Mean (μ)	₹2,000
Std. Deviation (σ)	₹400
Customer Spend (x)	₹3,200

Formula

$$z = (x - \mu) / \sigma$$

Calculation

Substituting the values:

$$z = (3200 - 2000) / 400 = 1200 / 400 = 3.0$$

Answer: $z = 3.0$

Interpretation

A z-score of 3 means the customer's spending is 3 standard deviations above the mean, which is quite far from typical behaviour. So yes, this spending is unusually high and can be treated as an outlier (a z-score above 2 is generally considered unusual).

A negative z-score would simply mean the customer spent less than the average ₹2,000. For a customer with $z = -1.5$, we can find the actual spend by rearranging the formula: $x = \mu + z\sigma = 2000 + (-1.5 \times 400) = ₹1,400$. So this customer spent about ₹1,400, which is below average but not extreme enough to be an outlier.

Z-scores are used in fraud detection, anomaly detection and customer segmentation, since they let us compare individual values against the average in a standard way, regardless of the original scale.

Case Conclusion: The customer's spending of ₹3,200 gives a z-score of 3.0, which is well outside the normal range and can be flagged as unusually high spending.

3. Case Study 7 – Delivery Time Analysis

Problem Statement

A company recorded 10 delivery times (in minutes): 25, 27, 28, 29, 30, 30, 31, 32, 34, 60. We need to find the mean, median, quartiles, IQR, and check for outliers.

Objective: To calculate the mean, median, quartiles and IQR of the delivery times, apply the $1.5 \times$ IQR rule, and decide whether the mean or the median better represents a typical delivery.

Given Data (sorted)

1	2	3	4	5	6	7	8	9	10
25	27	28	29	30	30	31	32	34	60

Mean and Median

Sum of all values = $25+27+28+29+30+30+31+32+34+60 = 326$

$$\text{Mean} = 326 / 10 = 32.6 \text{ minutes}$$

Since $n = 10$ is even, median = average of 5th and 6th values = $(30+30)/2$

$$\text{Median} = 30 \text{ minutes}$$

Quartiles and IQR

Lower half (25,27,28,29,30) $\rightarrow Q1 = 28$. Upper half (30,31,32,34,60) $\rightarrow Q3 = 32$.

$$\text{IQR} = Q3 - Q1 = 32 - 28 = 4$$

Outlier Check ($1.5 \times \text{IQR}$ Rule)

$$\text{Lower bound} = 28 - 1.5(4) = 22 \quad \text{Upper bound} = 32 + 1.5(4) = 38$$

All values lie inside [22, 38] except 60, which is greater than 38. So 60 minutes is an outlier.

Results

Mean	Median	Q1	Q3	IQR	Outlier
32.6	30	28	32	4	60

Interpretation

The outlier (60 minutes) pulls the mean up, so the mean (32.6) is higher than what most deliveries actually took. The median (30) is not affected by this extreme value, so it is a better representation of the typical delivery time here.

Case Conclusion: Out of 10 deliveries, one (60 minutes) is a clear outlier. Since it distorts the mean, the median is the fairer measure of the company's typical delivery time in this dataset.

4. Case Study 8 – Comparing Temperatures of Two Cities

Problem Statement

Daily temperatures ($^{\circ}\text{C}$) over 7 days for City A: 28,29,30,30,31,30,32 and City B: 20,25,30,35,30,25,45. We need to compare the mean and variance of the two cities.

Objective: To calculate the mean and variance of both cities' temperatures and to check which city has more variability and, therefore, a less predictable climate.

Formula

$$\text{Mean} = \Sigma x / n \quad \text{Variance } (\sigma^2) = \Sigma (x - \bar{x})^2 / n$$

City A

Sum = $28+29+30+30+31+30+32 = 210$, so Mean = $210/7 = 30^{\circ}\text{C}$

x	$x - \bar{x}$	$(x - \bar{x})^2$
28	-2	4
29	-1	1
30	0	0
30	0	0

x	$x - \bar{x}$	$(x - \bar{x})^2$
31	1	1
30	0	0
32	2	4

Sum of squares = $4+1+0+0+1+0+4 = 10$

$$\text{Variance} = 10/7 = 1.43, \text{ Std. Deviation} = \sqrt{1.43} = 1.20^{\circ}\text{C}$$

City B

Sum = $20+25+30+35+30+25+45 = 210$, so Mean = $210/7 = 30^{\circ}\text{C}$

x	$x - \bar{x}$	$(x - \bar{x})^2$
20	-10	100
25	-5	25
30	0	0
35	5	25
30	0	0
25	-5	25
45	15	225

Sum of squares = $100+25+0+25+0+25+225 = 400$

$$\text{Variance} = 400/7 = 57.14, \text{ Std. Deviation} = \sqrt{57.14} = 7.56^{\circ}\text{C}$$

Results

City	Mean	Variance	Std. Deviation
City A	30	1.43	1.20
City B	30	57.14	7.56

Interpretation

Both cities have the same mean (30°C), but City B has a much larger variance. This shows that a common mean does not mean the data behaves the same way — City A's temperature stays close to its average every day, while City B fluctuates a lot. So City A is more predictable/consistent, and City B is not.

Case Conclusion: City A (variance 1.43) is far more stable than City B (variance 57.14), even though both cities share the same average temperature of 30°C .

5. Case Study 9 – Data Center CPU Usage

Problem Statement

CPU usage (%) recorded over 10 intervals: 40, 42, 45, 45, 46, 48, 50, 52, 55, 90. Find the mean, median, mode, range, quartiles, IQR, outliers and skewness.

Objective: To summarise the CPU usage data using mean, median, mode and range, detect outliers using the IQR rule, and comment on the direction of skewness.

Mean, Median, Mode, Range

Sum = $40+42+45+45+46+48+50+52+55+90 = 513$

$$\text{Mean} = 513/10 = 51.3\%$$

$n = 10$ (even), so median = average of 5th and 6th values = $(46+48)/2$

$$\text{Median} = 47\%$$

45 is the only value that repeats (twice), so Mode = 45%. Range = Max – Min = $90 - 40 = 50\%$.

Quartiles, IQR and Outliers

Lower half (40,42,45,45,46) → Q1 = 45. Upper half (48,50,52,55,90) → Q3 = 52.

$$\text{IQR} = 52 - 45 = 7$$

$$\text{Lower bound} = 45 - 1.5(7) = 34.5 \quad \text{Upper bound} = 52 + 1.5(7) = 62.5$$

90 is greater than the upper bound of 62.5, so 90% is an outlier.

Skewness

Mean (51.3) is greater than Median (47), so the data is positively (right) skewed — the outlier of 90 pulls the mean up.

Results

Mean	Median	Mode	Range	Q1	Q3	IQR	Outlier
51.3	47	45	50	45	52	7	90

Interpretation

Most of the CPU readings stay close to 45–48%, but the reading of 90% is an unusual spike, correctly caught as an outlier by the IQR rule. This same spike is what makes the distribution positively skewed. In practice, a spike like this could mean a background process or an issue worth investigating.

Case Conclusion: The CPU usage data is mostly stable around 45-48%, with one clear outlier at 90% that also causes positive skewness in the distribution.

6. Case Study 10 – Comparing Two Datasets

Problem Statement

Two datasets represent errors of two prediction models. Dataset A: 10,20,30,40,50. Dataset B: 28,29,30,31,32. We need to compare their mean, variance and standard deviation.

Objective: To calculate the mean, variance and standard deviation of both datasets and decide which one represents a more consistent model, given that both have the same mean.

Dataset A

Sum = $10+20+30+40+50 = 150$, so Mean = $150/5 = 30$

x	$x - \bar{x}$	$(x - \bar{x})^2$
10	-20	400
20	-10	100
30	0	0
40	10	100
50	20	400

Sum of squares = $400+100+0+100+400 = 1000$

Variance = $1000/5 = 200$, Std. Deviation = $\sqrt{200} = 14.14$

Dataset B

Sum = $28+29+30+31+32 = 150$, so Mean = $150/5 = 30$

x	$x - \bar{x}$	$(x - \bar{x})^2$
28	-2	4
29	-1	1
30	0	0
31	1	1
32	2	4

Sum of squares = $4+1+0+1+4 = 10$

Variance = $10/5 = 2$, Std. Deviation = $\sqrt{2} = 1.41$

Results

Dataset	Mean	Variance	Std. Deviation
Dataset A	30	200	14.14
Dataset B	30	2	1.41

Interpretation

Both datasets have the same mean (30), but Dataset A has a much bigger spread. This means Dataset B's model is more consistent, since its errors stay close to 30 every time, while Dataset A's errors swing widely between 10 and 50. This shows that mean alone is not enough to judge two datasets — spread (variance/SD) has to be checked too.

Case Conclusion: Dataset B (SD = 1.41) is far more consistent than Dataset A (SD = 14.14), so a model producing errors like Dataset B would be considered more reliable.

7. Overall Conclusion

Doing these five case studies helped me understand descriptive statistics in a more practical way, rather than just as formulas. Case Study 7 and Case Study 9 showed how a single extreme value (a delayed delivery, a CPU spike) can pull the mean away from what is actually 'typical', and how the median stays more stable in such cases. The mode and range added a bit more detail about the CPU usage data.

Case Study 8 and Case Study 10 showed something I had not really thought about before — two datasets can have exactly the same mean and still behave very differently. Only by calculating the variance and standard deviation could we tell that City A and Dataset B were far more consistent than City B and Dataset A. This is why a mean should never be reported alone.

The IQR and the $1.5 \times \text{IQR}$ rule, used in Case Study 7 and Case Study 9, gave a clear and repeatable way to detect outliers instead of just guessing by looking at the numbers. Case Study 6 tied things together with the z-score, which standardises any value so it can be compared against the mean in terms of standard deviations — this is what is actually used in things like fraud detection.

Overall, this assignment made it clear that mean, median, mode, variance, standard deviation, quartiles, IQR and z-score are not separate topics but a connected set of tools that Data Scientists use to understand and clean data before doing any further analysis or building any model.